



# Box office prediction based on microblog



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## ABSTRACT

As the importance and popularity of online social media has become more obvious, there are more researches aiming at making use of information from them. One important topic of this is predicting the future with social media. This paper focuses on predicting box offices using microblog. Compared with previous work which makes use of the count of related microblogs simply, the information from social media has been utilized more deeply in this paper. Two sets of features have been extracted: count based features and content based features. For the former, the information in the aspect of users, which decrease the influence of garbage microblogs, has been exploited. For content based features, a new box office oriented semantic classification method has been provided to make the features more relative with box offices. Meanwhile, more complex machine learning models such as SVM and neural network have been applied to the prediction method. Our prediction model is more accurate and reliable. With our prediction method, the data in Tencent microblog has been utilized to predict box offices of certain movies in China. With the results, the strength of our method and predictive power of online social media can be completely demonstrated.

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## 1. Introduction

With the development of World Wide Web, people would like to spend more time on the Internet as well as the social media online. Because of their convenience and ease of use, social media, such as microblog, twitter and facebook, have become platforms for recording experiences, sharing opinions and communications. Due to social media's character of high popularity, the opinions that people express on social media can, to some extent, represent the opinions of most people in the real world. From this respect, the social media can be precious resources of information for researchers to observe, understand, and even explore the events and laws of the real world.

This paper studies the power of microblog to predict events and outcomes in real world. More specifically, our goal is to predict the box office of movies using the information we extract from microblog. Moreover, our work demonstrates the predictive power of microblog as well as online social media. Besides the box office, our method can also be used to predict other outcomes in the real world such as sales volume and stock.

Making good use of information from online social media is an important issue in this age. One aspect of this is predicting outcomes and data in real world using social media. There are many researches on it. However, no one can be sure to say the prediction methods based on social media are feasible and accurate to predict all the outcomes. Around this problem, researchers can be divided into two categories. Some researchers believe that information from social media can be used to predict data from real world and provide some practical prediction methods. On the contrary, through their experiments and comparison with traditional methods, other researchers demonstrate that prediction methods based on social media are not appropriate to predict some outcomes and events.

People try various methods to extract information from social media and construct different prediction models to predict various things. For example, Guille and others built a predictive model for the information diffusion in online social networks (Guille & Hacid, 2012). Nguyen and others presented a strategy of building statistical models from the social media dynamics to predict collective sentiment dynamics (Nguyen, Wu, Chan, Peng, & Zhang, 2012). Gupta and others predicted the credibility of information in a tweet (Gupta & Kumaraguru, 2012). Besides these, there are lots of work about similar problems (De Choudhury, 2009; Gilbert & Karahalios, 2009; Hong, Dan, & Davison, 2011; Huang, Chen, Luo, & Lee, 2012; Lerman & Hogg, 2010, 2012; Lerman, Intagorn, Kang, & Ghosh, 2012; Qi, Qu, & Tan, 2012; Rosenthal & McKeown, 2011; Tang & Liu, 2009). However, above work is predicting information

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of social media themselves but not very relative with the real world. There is no direct comparison between our work and their methods. On the other hand, there is also much work on predicting the real outcomes and events with social media. Asur and others used data from twitter and built a model based on the rate at which tweets are created about particular movies as well as the sentiment polarity of these tweets to predict the box offices of the movies (Asur & Huberman, 2010). In their paper, they provided a general prediction model for social media. However, they did not demonstrate the effectiveness of this model through experiments and their model is so simple that can be influenced by various factors. Liu and others focused on content information of blog (Liu, Huang, An, & Yu, 2007). They used sentiment PLSA to find the hidden sentiment factors in blogs and built an autoregressive sentiment-aware model to predict product sales performance. They concentrated on the opinions and sentiments in the blogs. However, because their dataset is from blog, which are long-text social media, they could not make use of features based blogs' count. And Bar-Haim and others predicted stock price movement using stock microblogs (Bar-Haim, Dinur, Feldman, Fresko, & Goldstein, 2011). Their prediction method is mainly finding expert investors and collecting experts' opinions. Besides, there are other researchers who predicted movie ratings and stock market level with twitter data (Mao, Wang, Wei, & Liu, 2012; Oghina, Breuss, Tsagkias, & de Rijke, 2012). Some of methods above are not general enough while others are not accurate enough. Because of different characters of different social media, it is not appropriate to apply the same prediction method to different social media.

Contrarily, other people demonstrated that prediction methods using social media cannot be applied to some problems. For instance, Skoric and others tried to predict election result in Singapore with twitter (Skoric, Poor, Achananuparp, Lim, & Jiang, 2012). However, they found that the predictive power is so weak in this problem because there are many other complex factors. Moreover, Gayo-Avello suggested that the power to predict outcomes based on twitter data is greatly exaggerated, especially for political elections (Gayo-Avello, 2011). Actually, prediction methods based on social media are not so good at predicting outcomes which are influenced by lots of factors and not so relative with social media.

Compared with previous work above, the information has been exploited more deeply and variously in this paper. The improvement of methods in the respects of amount and content of microblog can immensely optimize the prediction result. Different from previous researches, our method mix the count based information and the content based information. Furthermore, instead of linear prediction model, the machine learning models like neural network and SVM which are more complex to predict the box offices have been chosen. The results of the experiments show that the outcomes can be predicted more precisely and reliably after applying our method.

The rest of this paper is arranged as follow. The problem we would like to solve will be defined in Section 2. Besides this, the framework of our method will be provided in this section. Section 3 describes the feature extraction methods and prediction model specifically. Section 4 presents our experiments to demonstrate the predictive power of the methods and discussion about them. At last, the result of the paper will be concluded and the future work will be presented in Section 5.

## 2. Problem definition and framework

### 2.1. Problem definition

As we said in Section 1, our goal is to demonstrate the predictive power of microblog. To achieve this, a more specific problem

should be defined. Therefore, our focus turns to predicting the box offices of movies.

Assume  $x_i$  to be the information, which is extracted from microblog which is published during the days from  $t_{i-1}$  to  $t_i$ . Moreover, it is assumed that  $y_i$  is the box office of certain movie during days from  $t_i$  to  $t_{i+1}$ . Without loss of generality, it is assumed that  $t_i - t_{i-1} = 7$ . That is to say, the information and box office is split into some periods whose lengths are 7 days. Besides, it is assumed that  $t_1$  is the day when the movie was released. So the problem is to predict  $y_i$  using  $x_1, x_2, \dots, x_i$  as well as  $y_1, y_2, \dots, y_{i-1}$ . In other word, the prediction of

$$y_i = g(x_1, \dots, x_i, y_1, \dots, y_{i-1}) \quad (1)$$

will be our major work. Here, it is confirmed that  $i = 3$ , so our problem is to predict

$$y_i = g(x_1, x_2, x_3, y_1, y_2) \quad (2)$$

In other word, the box office of the movie on the 3rd week will be predicted using the microblog information from 1 week before it was released to 2 weeks after it was released as well as the box offices of first 2 weeks.

This problem can be divided into 2 sections. One of them is to choose a proper prediction model. The other is to find features to accurately and entirely represent the information of microblog.

### 2.2. Framework

To solve the problem above, our prediction method can be summarized as Fig. 1.

The microblog website is crawled with the API of it at first to get the microblogs which are relative with the movies. After this, the information such as users, contents and comments is extracted from the microblogs. With the information, our content based method and count based method can be used to get the features and these features are mixed. The mixed features are applied to the machine learning models to build prediction models. The prediction models can be trained by real data. To predict box office of a new movie, we can extract the features as the input of the trained prediction model. The output of the model is the box office of the movie which we predict.

In following sections, the feature extraction methods are going to be described. After that, the prediction models will be provided to solve the problem.

## 3. Proposed method

### 3.1. Feature extraction method

#### 3.1.1. Overview

As we can see in the last section, one of the most important parts of our work is to represent the information we can get from microblog as an input of our prediction model. Lots of previous work about prediction using microblog divides this task into two separate sections. One section is extracting the information based on the amount of microblogs, comments and retweets. These features, which named as count based features, are independent on the contents of the microblogs. The other section is making use of the information about the contents of the microblogs such as the amount of microblogs with positive sentiment and the ratio of microblogs with positive sentiment and negative sentiment. Previous work shows these features can be used to predict the box office with good performance. However, there may be interaction effect between the two kinds of features so it may be better to mix the two. In our work, not only count based features but also content based features are improved to make use of the microblog

information more deeply. Furthermore, we provide new features which have both count based and content based information. It can be shown that the new features are more relative with the box office.

The rest of this section will be arranged as follow. Our count based and content based feature will be provided in Sections 3.1.2 and 3.1.3. At Section 3.1.4, the mixed feature will be showed and some discussion and improvement will be presented.

### 3.1.2. Count based feature extraction

Before describing the method, some characters of microblog and the behaviors of the users need to be analyzed. Through the analysis, following conclusions can be reached.

First of all, different users have different effect on prediction model. The users who are followed by more people have more value. Their opinions can influence more people. Meanwhile, the users whose microblogs are commented and retweeted by more people can also have more values because their opinions can represent the opinions of more people. In this paper, this kind of users is named as “experts”.

Besides, there are a group of people who are not the real users of microblog. These people are named as “garbage users”. Their purpose to use microblog is to put advertisements on microblog. There are little real users who pay attention to their microblogs. During the prediction of box office, the garbage users and their microblogs can be seen as noise. By analyzing their behaviors, their characters can be summarized as following.

1. The amount of microblogs they posted is enormous.
2. The amounts of comments and retweets of their microblogs are little.
3. They follow a lot of people but few people follow them.

Based on above, existing methods can be improved and our proposed count based feature can be acquired.

In previous work, the method extracting count based feature is based on the amount of microblogs or the data relative with microblogs. However, considering the existence of garbage users, the performance of the prediction model would be poor. Besides, as we conclude above, the influences of different microblogs are different, and the difference is related to the influence of users, so it is more scientific to extract the feature based on the information of users. In this section, a new count based feature is proposed based on users' influences.

Before describing our feature, the influences of users in our problem need to be redefined. It is assumed that an “expert” is a kind of user who can be agreed by many people. The experts' opinions can represent a large amount people's opinions. It is assume that experts we describe above can also have much influence on others. This kind of influence is named as “authority” and the criterion to authority is called “authority score (AS)”.

$$AS = \frac{\sum_{m=1}^{i-1} (count_i + mcount_i + 1)}{m} \quad (3)$$

where  $m$  is the amount of microblogs the user posted during the time,  $count_i$  is the amount of retweets of the user's  $i$ th microblog and  $mcount_i$  denotes the amount of comments of this microblog.

Authority score reflects the degree to which the user was concerned. It uses the information from microblogs to evaluate the authority of users. As we can see, for a certain person, the authority score may be different at different time, which reflect the dynamic character of authority.

The quality and quantity of the users during the time is utilized to reflect how a certain movie is concerned. So the feature can be represented as follow.

$$x_t = \sum_{users} AS_j = \sum_{users} \frac{\sum_{m=1}^{i-1} (count_i + mcount_i + 1)}{m} \quad (4)$$

where  $AS_j$  is the  $j$ th user's authority score.

Therefore, the count based feature extraction method can be described as Algorithm 1.

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#### Algorithm 1. Count based feature extraction

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line size =

**Require:**

The set of all the microblogs related with the movie,  $M$ ;

The set of all the users of these microblogs,  $U$ ;

The timestamp of the 3 time periods,  $T_1, T_2, T_3$ ;

**Ensure:**

The count based feature,  $x_k$ .

1:  $k \leftarrow 0$ ;

2: **for**  $k \leq 3$  **do**

3:  $x_k \leftarrow 0$ ;

4: initiate  $m_j$  and  $AS_j$  of  $user_j$  in  $U$  to be 0;

5: **for each** microblog  $m_i$  in  $M$  **do**

6:  $t_i \leftarrow$  the post timestamp of  $m_i$ ;

7: **if**  $t_i$  between  $T_k - 864000 * 7$  and  $T_k$  **then**

8:  $AS_j \leftarrow count_i + mcount_i + 1$ ;

9:  $m_j \leftarrow m_j + 1$ ;

10: **end if**

11: **end for**

12: **for each**  $user_j$  in  $U$

13: **if**  $m_j > 0$  **then**

14:  $AS_j \leftarrow AS_j \div m_j$ ;

15:  $x_k \leftarrow x_k + AS_j$ ;

16: **end if**

17: **end for**

18:  $k \leftarrow k + 1$ ;

19: **end for**

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Note that 864,000 is amount of seconds in one day and  $AS_j$  is the authority score of the  $user_j$ . In Algorithm 1, the authority scores of different users in every week are calculated. After that, the total authority scores of this week are added up and this sum is one of the features.

Compared with previous count based features, this feature decreases the noise from the garbage users and count on the differences belonged to the users as well as the microblogs. Our experiments in Section 4 show that this feature is more relative with the box office.

### 3.1.3. Content based feature extraction

To make full use of the information on microblog, the content of the microblogs has to be considered. Our focus is the opinions the users express in their comments on the movies and microblogs. In this section, a new method is provided to use the information of the comments.

Most of previous works use sentiment polarities of the microblogs directly as the feature of prediction model. These methods are based on the assumption like this.

**Assumption 1.** If there are more positive microblogs about the movie, the box office of the movie will be better.

However, there is no necessary connection between users' sentiments and movies' box offices. Therefore, the assumption is not reliable. Indeed, people would like to see the movies that the users speak highly of. However, this does not lead to good box office.

To make it more reliable, the assumption is modified like this.

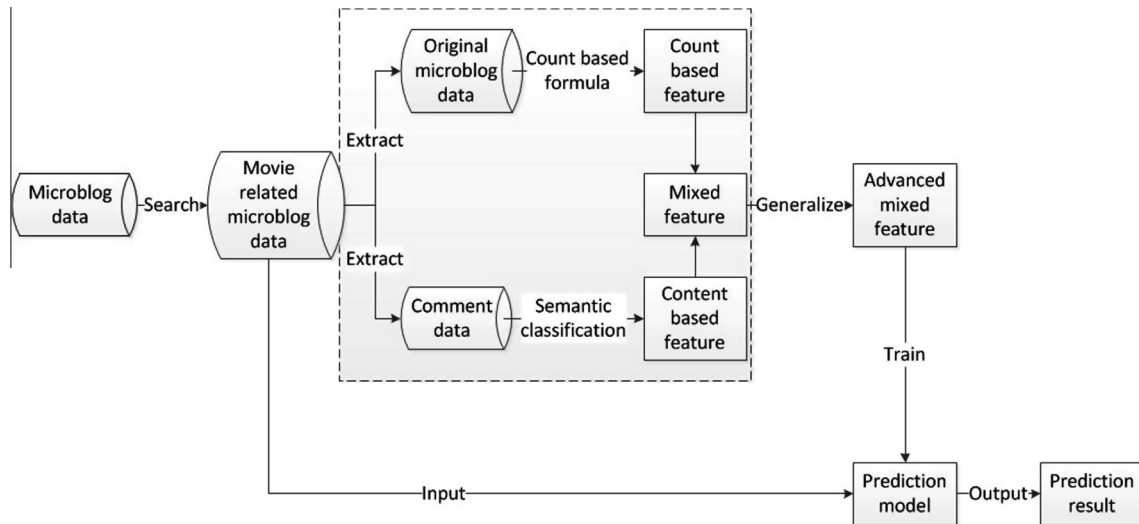


Fig. 1. Framework of microblog prediction method.

**Assumption 2.** If there are more microblogs whose contents are relative with the growth of box office, the box office of the movie will be better.

Based on the assumption above, the comments are divided into 3 classes: beneficial, harmful and neutral.

1. Beneficial: when a content of a microblog may lead to increment of the box office, the microblog is defined as beneficial. For example,

**Example 1.** This movie is wonderful, you should have a try!!! On the one hand, the sentence expresses a kind of positive opinion; on the other hand, it may let more the user's friends to go to see the movie. Meanwhile, there are some microblogs which don't contain positive opinions but encourage people to go to the movie like.

**Example 2.** I'd like to see it, who can go with me? Come on!

2. Harmful: contrary to beneficial microblogs, harmful microblogs imply the probability of decrease of the box office, such as

**Example 3.** It doesn't deserve to see ...

3. Neutral: The microblogs which are not related with the movie or don't contain any opinion are considered as neutral microblogs. For example,

**Example 4.** I'd love to have breakfast with you tomorrow.

Through our instances and analysis above, it is obvious that our new semantic classification method is more relative with the box office. As the experiments say in Section 4, this method can make the prediction model more accurate.

The microblogs we classify are in Chinese, so our classification method aims at Chinese corpus. Using the API of Tencent microblogs, the microblogs of which topic are related to any of 17 movies we selected are obtained and imported into database. Afterwards, almost 3000 microblogs are labeled which comprise our training corpus. Our classification method can be split into 4 sections: semantic label, preprocessing, text feature extraction, classifier training.

1. *Semantic label*: As we mention above, our label rules differ from the rules enacted in previous paper. The microblogs which contribute positively to the box office, although it may not appear to be positive, are marked as beneficial. Meanwhile the microblogs which impact negatively on the box office is labeled as harmful. And the rest are marked as neutral.

2. *Preprocessing*: By using ICTCLAS (ICTCLAS), all the microblogs which are obtained from microblogs have been at first segmented. In our sentiment analysis model, a document is represented as a bag of words. The words of which the parts of speech are numeral, auxiliary word and stop word are neglected. The remaining words are regarded as candidate feature.

3. *Text feature extraction*: Considering the variety of the sentence structure, 5 basic features are created which will be combined to form the final feature in the following part. Each type of basic feature is defined as follows.

- (a) *Word unigram*: Unigram patterns which appear in at least 2 distinct sentences in the corpus. Since a Chinese text comprises of large number of words, not characters, the unigram patterns refer to words in our following experiments.
- (b) *Adjective + verb + adverb*: Every adjective, verb, adverb which appear in the corpus.
- (c) *Sentimental words*: Unigram patterns which appear in our sentimental words dictionary.
- (d) *Frequent word subsequence*: Frequent word subsequence patterns whose sizes are 5 or more.
- (e) *Unigram or subsequence filtered by chi-square test*: In our sentimental analysis model, the chi-square test is applied to analyze the relationship between feature words and beneficial, harmful, neutral tweets. Feature words whose relativity exceeds the setting threshold will be selected as the updated feature words. The calculation of relativity is conducted as follows. Relativity of feature words is separated into three parts: patterns and beneficial tweets, patterns and harmful tweets, patterns and neutral tweets. Only if one part of relativity exceeds the threshold which is set as 2 can be taken into consideration.

4. *Classifier training*: Support vector machines (SVMs) with linear-kernel are chosen as the classifier to train and predict the content of microblogs. The parameter C which applied in the linear-kernel is set as 1. Cross-validation test is carried out to select the best classifier. Combining the 5 basic features, several mixed features are formed. And the final features, more accurate than the rest features using SVM classifier, are confirmed as unigram filtered by chi-square test plus subsequence.

The classification process can be shown in Fig. 2.

Using the method above, all the microblogs are labeled by their box office oriented semantic. In next section, the discussion is

focused on how to use this information and how to mix information from contents and count of microblogs.

### 3.1.4. Content and count mixed formula

Before the method to make use of the content information is described, relation between content based information and count based information should be paid attention to.

In Section 3.1.2, we described authority score (AS) to measure a certain user's influence on the prediction model. However, there are some factors which are not considered about this criterion such as the opinion of the user's microblogs. In this section, our previous assumption is extended as well as authority score.

It is defined that the expert to be the users who can be agreed by many people. Therefore, it is necessary to add the content based information to the authority score. So our advanced authority score is like this,

$$AS = \frac{\sum_{m=1}^{i-1} (count_i + 2 * benemcount_i - harmmcount_i + neumcount_i + 1)}{m} \quad (5)$$

where  $benemcount_i$ ,  $harmmcount_i$  and  $neumcount_i$  are the amounts of microblogs whose box oriented semantics are beneficial, harmful or neutral.

Advanced authority score mixes count based information as well as content based information and gives us a more proper way to measure the value of a user for the prediction model. It is a kind of scoring scheme.

Adding advanced authority score to our previous feature, formula which mixed count based information and content based information can be acquired.

$$x_t = \sum_{users} AS_j = \sum_{users} \frac{\sum_{m=1}^{i-1} (count_i + 2 * benemcount_i - harmmcount_i + neumcount_i + 1)}{m} \quad (6)$$

In the mixed feature extraction method, the semantic classification method can be seen as one subprocess of the count based feature extraction method. Essentially, the value of formula is the sum of AS of the users during the time. It considers both the quality and quantity of the users. Its effectiveness will be estimated through experiments in the Section 4.

### 3.1.5. Advanced formula

As we said in previous section, authority score is based on a kind of scoring scheme and the score is set artificially. However, to measure the authority of a microblog or a user, the weights of different classes of semantics are different. In other word, it is not reasonable to offer a fixed score to a certain semantic. Therefore, our previous formula is generalized as follow.

$$x_t = \sum_{users} AS_j = \sum_{users} \frac{\sum_{m=1}^{i-1} (\alpha * (count_i + 1) + \beta * benemcount_i + \gamma * harmmcount_i + \delta * neumcount_i)}{m} \quad (7)$$

where  $\alpha, \beta, \gamma, \delta$  are the weights of different comments with different semantics as well as retweets. To make it more clearly, the feature AS is represented as a vector.

$$AS = \begin{pmatrix} \sum_{users} \frac{\sum_{m=1}^{i-1} count_i + 1}{m} \\ \sum_{users} \frac{\sum_{m=1}^{i-1} benemcount_i}{m} \\ \sum_{users} \frac{\sum_{m=1}^{i-1} harmmcount_i}{m} \\ \sum_{users} \frac{\sum_{m=1}^{i-1} neumcount_i}{m} \end{pmatrix}$$

Similarly, the weights are expressed as a vector.

$$W = \begin{pmatrix} \alpha \\ \beta \\ \gamma \\ \delta \end{pmatrix}$$

So the vector  $X_t$  is as follow,

$$X_t = W^T AS \quad (8)$$

The weight can be trained by the data we get from the microblogs and using formula above to get the feature.

The proposed feature extraction methods are described above. Through our analysis, compared with traditional methods, our features are more proper and reasonable. Their performance will be showed in practical application in Section 4. In next section, prediction models will be improved.

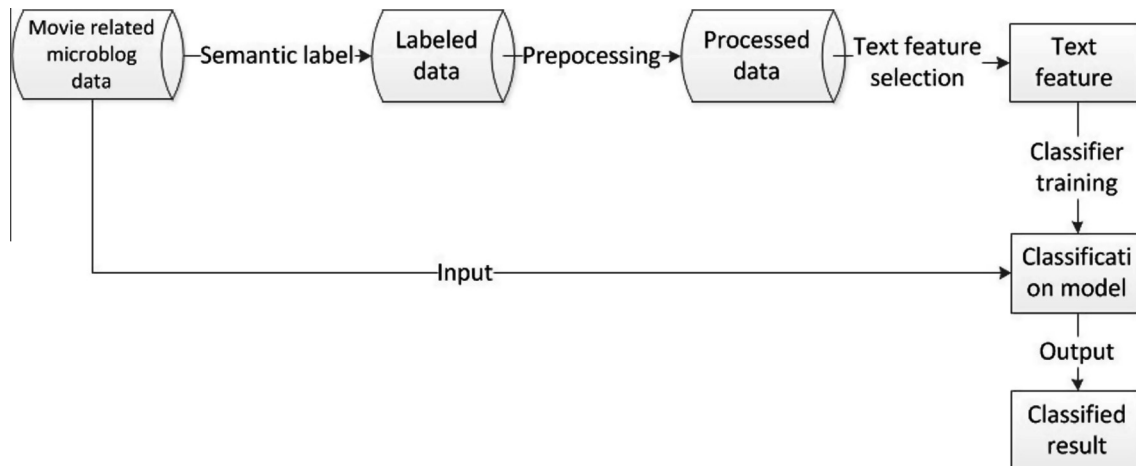


Fig. 2. Semantic classification process.



### 3.2. Prediction model

Our feature extraction methods are described in Section 3.1. In this section, our focus is to determining an appropriate prediction model for our problem.

In essence, our microblog prediction model is a regression model. As we said in Section 2, the input of our model is as follow.

$$AS = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ y_1 \\ y_2 \end{pmatrix}$$

Here,  $x_i$  is the feature we extract from microblogs in the  $i$ th week and  $y_i$  is the box office of the movie in the  $i$ th week.

The output of the prediction model is  $y_3$ .

Our next task is to select a prediction model to fit the relationship between our input and output. In this paper, 3 existing regression models are tried and their performances are tested in our experiments.

#### 1. Linear model

In many previous works, people assume the relationship between their prediction objects, like box offices and stocks, and their features are linear. Based on this, they use linear models to predict their objects. In our problem, we use their ideas for reference.

Our model is represented as follow.

$$y_3 = g(i) + \epsilon = W^T + \epsilon \quad (9)$$

where  $W = (w_1, w_2, w_3, w_4, w_5)^T$  is the vector of the weights and  $\epsilon$  is a constant here.

To make the model simpler, it is modified as follow.

$$y_3 = W'^T I' \quad (10)$$

And  $W' = (w_1, w_2, w_3, w_4, w_5, \epsilon)^T$ ,  $I' = (x_1, x_2, x_3, y_1, y_2, 1)^T$ .

Using the train data we collect from different movies, the value of the weights can be confirmed. And then, this model are used to predict the box office of a new movie.

#### 2. Nonlinear model

However, the relationship may be more complex. The assumption above is unreliable and unsubstantial. A more expressive model is needed to represent it. Therefore, in this paper, some powerful nonlinear models are tried to predict the box office. Here, neutral network model and support vector machine (SVM) are selected to be our prediction models. These models both are robust regression models and are able to fit most of the nonlinear functions. In this aspect, the prediction results will be accurate enough using these models if the box offices and microblogs have certain relationship.

Our 3 choices of the prediction models are described. The linear model is simple but the nonlinear models may be more accurate. In next section, we will compare these models and demonstrate this speculation through our experiments.

## 4. Experiment

### 4.1. Experiment setting

To demonstrate the predictive power of microblog, predicting box offices of movies using microblogs will be focused. Real box office and microblog data are collected to train our model and test the performance of our models.

#### 1. Box office data

Our box office data come from the Entgroup box office website ([Entgroup Box Office](#)). To test our methods more comprehensively, different types of movies are chosen which are from different countries. To match our microblog data, the Chinese box offices are chosen. The data we collected are first 3 weeks after the movies are released. This is fit our problem we defined in Section 2.

Noting that our box office data are divided by week but the movies are released on different days in every week, the box office data of the first week is needed to modify. The method we use to modify the data is as follow.

$$y'_i = \frac{y_i * 7}{8 - m} \quad (11)$$

Here the movie is released on the  $m$ th day in the first week.

#### 2. Microblog data

The microblog dataset we used is from [Tencent Microblog](#). The data are collected through its search API. The search keywords are titles of the movies. However, there are some movies whose names are words people use every day. To make our results more accurate, some punctuation like quotation marks are added to these titles. To calculate values of our formulas, following information is acquired using the API.

- Contents of the microblogs
- Amounts of the retweets and comments of the microblogs
- Authors of the microblogs
- Time when the microblogs were posted

Using these microblogs, the contents of their comments are crawled through the API. The microblogs which were posted from one day before the movies were released to two days after that are collected. Microblog data are divided weekly to match our box office data.

Collected 24 movies' box offices are collected and 120,413 microblogs about these movies from 68,269 users. Because the inputs of our prediction models are in different dimensions, it is important to regularize these values as follow.

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (12)$$

For the machine learning models of the methods, we use [Lib-SVM](#) to implement the SVM model and [Knime](#) to implement the neutral network model. The linear-kernel of SVM are chosen and the parameter  $C$  which applied in the linear-kernel is set as 1. As for the neutral network model, we set the number of hidden layers as 1 and number of neurons per layer as 10.

The arrangement of our experiments is as follows. At first, the strength of our feature extraction methods is focused on. Our formulas will be compared with previous methods from the aspects of count based and content based features. And then, the performances of our prediction models will be tested. Besides these, the microblog prediction methods will be compared with traditional prediction methods to demonstrate the prediction power of microblog. In the end, some discussions will be made about the results of above experiments.

### 4.2. Comparison among count based methods

In this section, the methods without content information are focused on. According to previous work, following features are chosen to compare with our methods.

- Tweet-rate.** To some extent, the post rate of microblogs referring to a certain movie is relative with its box office. Meanwhile, this is one of the most direct features of the microblogs. Lots of

**Table 1**  
Correlations of count based methods during different time.

| Feature       | Correlation coefficient |
|---------------|-------------------------|
| Tweet-rate    | −0.017224749            |
| Comment-rate  | 0.259208978             |
| Formula       | 0.298916302             |
| Mixed formula | <b>0.310198689</b>      |

**Table 2**  
Correlations of count based methods among different movies.

| Feature       | Correlation coefficient |
|---------------|-------------------------|
| Tweet-rate    | −0.06654                |
| Comment-rate  | −0.13632                |
| Formula       | 0.050495                |
| Mixed formula | <b>0.103163</b>         |

previous work pays attention to this feature and uses it to predict many events or real data. Because the box offices and features are divided weekly, in our experiments, tweet-rate is defined as the number of microblogs referring to a movie per week.

2. *Comment-rate*. The amount of comments about a movie during a period of time reflects its degree of attentions. In this aspect, the comment-rate can be a feature of our prediction model. In our experiments, comment-rate is considered as the amount of comments of the microblogs about the movie during the week.

Besides these, we select two of our features to demonstrate the predictive power of our methods.

1. Formula without content information.
2. Content and count mixed formula

Before accuracies of different prediction models with these features are evaluated, the degrees of correlation between the features and box offices are compared. As first, our focus is on

the correlations between features and box offices in different time. The results are as [Table 1](#).

And then, the correlations of different movies are compared. We calculated correlation coefficients between the output of different prediction models (box offices in the 3rd week) and the inputs of them. The results are shown as [Table 2](#).

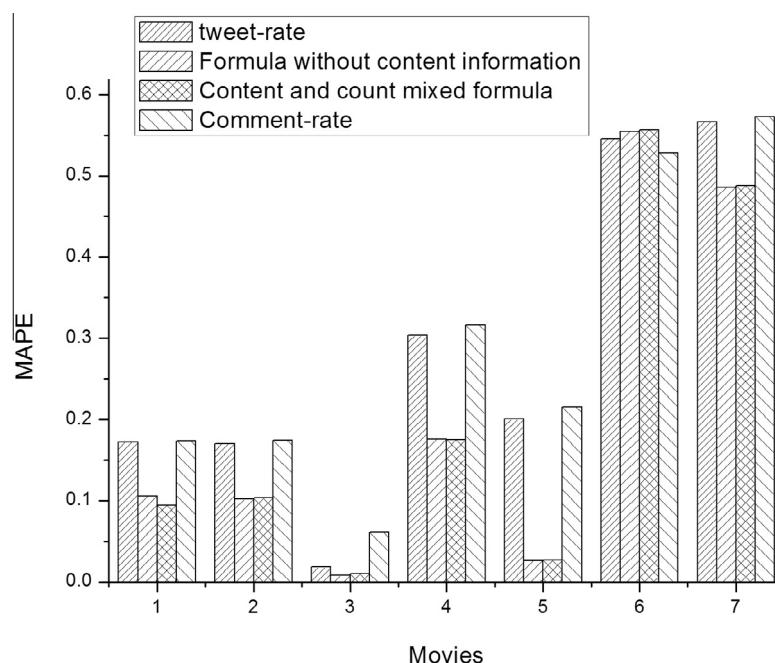
After these, these features and SVM prediction model are applied to predict the box offices. 17 movies are chosen randomly as train data and 7 movies as test data. Our evaluation criteria are the mean absolute percentage error (MAPE) of outputs from the prediction models. The MAPE of every test data are as [Fig. 3](#).

And the average values of these data are calculated. The result is shown in [Table 3](#).

From the experiment results above, the strength of our feature extraction methods can be demonstrated. On the one hand, our features are more relative with the box office. On the other hand, the MAPE are smaller using our features. Meanwhile, mix formula is better than formula without content information. This fits our assumption in [Section 3](#). Therefore our analysis about prediction using microblog is correct and rational. Compared with previous features, the values of our formulas can make the prediction models more accurate.

Besides above, some other conclusions are found.

1. Comment-rate is not a good choice. The amount of comments is more relative with their original microblogs than box offices. In this aspect, it is more appropriate to make use of content information in comments, not their amount.
2. The accuracies of our models about different movies are different. Through our observation, it is observed that when predicting box offices of domestic movies, our models have better performance. Actually, domestic movies are more relative with microblog users. And information from microblogs of them is influenced by the box offices more observably. This reflects one character of microblog prediction methods. These methods can predict some real data perfectly, such as box office and book sales volume. However, for some other problem like predicting the result of an election, the microblog prediction methods cannot have good performances.



**Fig. 3.** MAPE of count based methods for different movies.

**Table 3**  
Average MAPE of count based methods.

| Feature       | MAPE               |
|---------------|--------------------|
| Tweet-rate    | 0.282515906        |
| Comment-rate  | 0.29169102         |
| Formula       | 0.208340126        |
| Mixed formula | <b>0.207720674</b> |

**Table 4**  
Correlations of content based methods.

| Feature   | Correlation coefficient |
|-----------|-------------------------|
| Semantic  | <b>0.310198689</b>      |
| Sentiment | 0.30025362              |

#### 4.3. Comparison among content based methods

In this section, we consider the performances of our content based features and compare them with other content based features.

To show the power of our box office oriented semantic features, it is compared with traditional sentiment features. As we said in Section 3, our semantic classification method can be applied to traditional sentiment classification. Only the label rule needs to be modified. When the comments according to their sentiment polarities are labeled, our classification method turns to be traditional sentiment classification method. To make our experiments more convincing, the same corpus and classification model are applied to the two content based feature extraction methods. We add them to our content and count mixed formula and compare the performances of the mixed features.

Similar with Section 4.2, their correlations with the box offices are compared firstly. The results are as Table 4.

And then, the features are put into our prediction models. We still use SVM prediction model and select 17 movies as train data. The MAPE of these models are as Fig. 4 and Table 5.

As we can see in the results, our box office oriented semantic features are more relative with traditional sentiment features. The prediction model is more accurate using our methods.

**Table 5**  
Average MAPE of content based methods.

| Feature   | MAPE               |
|-----------|--------------------|
| Semantic  | <b>0.207720674</b> |
| Sentiment | 0.207828709        |

Compared with formula without content information, the performances of the two mixed formulas are both better. This demonstrates that adding content based information to the count based features is necessary to make the prediction models better. Meanwhile, this can reflect the rationality of our method to mix the two kinds of information.

#### 4.4. Comparison among prediction models

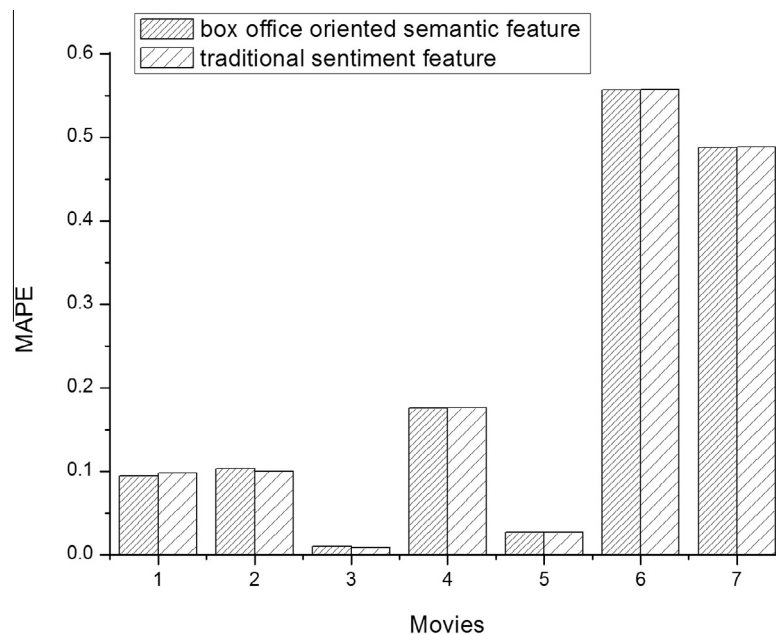
In Sections 4.3 and 4.4, we evaluate our feature extraction methods and demonstrate the predictive powers of our methods. Our next task is to select a better regression model for our problem. In this section, the three models referred in Section 3 are evaluated.

As Sections 4.3 and 4.4 described, our mixed formula with box office oriented semantic information has the best performance. In this section, these features are used as a part of the inputs. 17 movies are still selected randomly as train data and MAPE is used as criteria. The results are as Fig. 5 and Table 6.

In the aspect of accuracy, simple linear regression model is not appropriate to be the prediction model. More complex nonlinear models are more accurate and have smaller errors. Meanwhile, neural network model is better than SVM model.

#### 4.5. Comparison with traditional prediction methods

In this paper, our goal is to demonstrate the predictive power of microblog. It is necessary to compare our microblog prediction methods with traditional prediction methods. Actually, there are various traditional prediction methods which can be applied to our problem. However, these methods are not key points of this paper. So we choose one of the traditional methods which only remove the microblog information from our features, that is to say



**Fig. 4.** MAPE of content based methods for different movies.



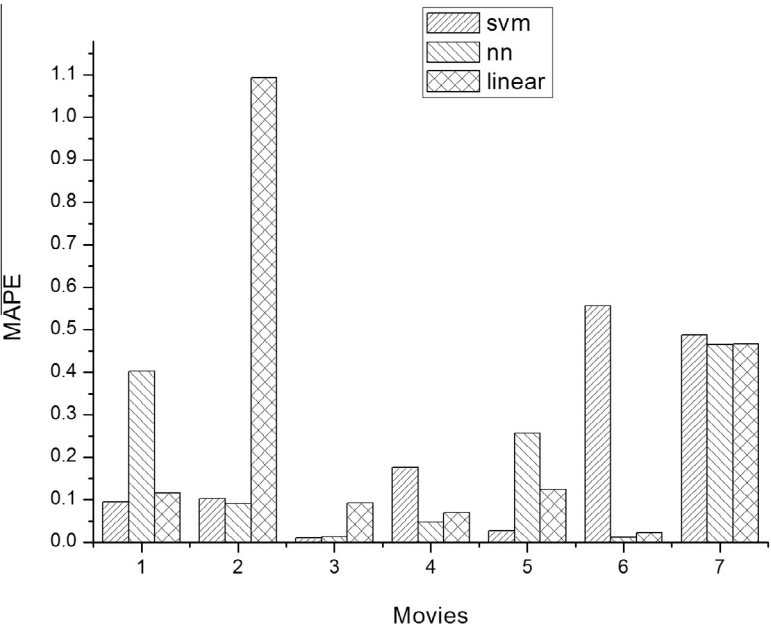


Fig. 5. MAPE of different prediction models for different movies.

Table 6  
Average MAPE of different prediction models.

| Prediction model | MAPE               |
|------------------|--------------------|
| Linear           | 0.283259444        |
| SVM              | 0.207720674        |
| Neutral network  | <b>0.183766872</b> |

box offices in the first two weeks are used to predict the box offices in the third week. In other word, the input is modified like this.

$I = (y_1, y_2)^T$

And the output is still  $y_3$ .

The same train data and the same regression model (neutral network model) are used here. The MAPE of the two outputs are compared. The results are shown in Fig. 6 and Table 7.

The gap on the accuracy is obvious between the two. This to some extent reflects the predictive power of microblog. Adding microblog information improves the performances of traditional prediction models observably. However, as we concluded in Section 4.2, microblog prediction methods may not be able to have good performances on some kinds of problems. It is one of our future work to find out microblog prediction methods can be applied to predict what kind of outcomes perfectly.

4.6. Discussion

Through the experiments above, the prediction power of our methods and models are demonstrated. From the aspect of feature extraction, our methods are more accurate and more related to the box offices. As for the prediction model, our machine learning models are complex and expressive.

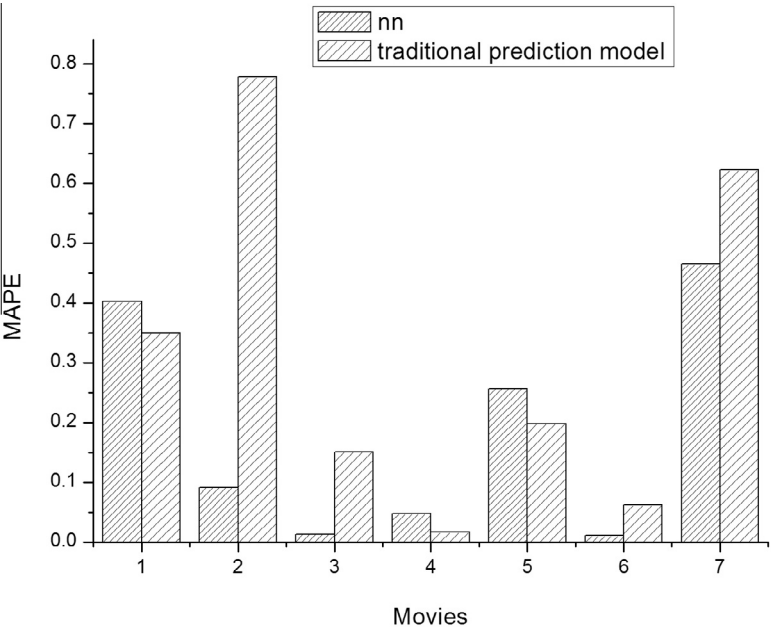


Fig. 6. MAPE of different prediction methods for different movies.

**Table 7**

Average MAPE of different prediction methods.

| Prediction method             | MAPE               |
|-------------------------------|--------------------|
| Microblog prediction method   | <b>0.183766872</b> |
| Traditional prediction method | 0.311018874        |

From the macro point of view, the prediction method is modularized. The method can be divided into three separate modules clearly, content analysis module, feature generation module as well as prediction module. Each of the three can be enhanced independently and replaced by other better methods. This character also provides the possibility of cross-language social media prediction. The model can predict box office with social media from various countries if only we use different semantic classification methods.

Besides, some important characters of microblog prediction methods are found through the experiments. They can help us to use microblog prediction methods to appropriate problems and make good use of these methods.

## 5. Conclusion and future work

Is it an appropriate way to predict real world outputs using microblog information? To some extent, an answer for this question is provided in this paper. We demonstrated the predictive power of microblog by offering more reliable and accurate methods to solve this problem.

Previous methods are improve in two aspects, feature extraction method and prediction model. Based on previous works, new count based feature extraction method and content based feature extraction method are provided. Our count based method extracts the features from the users' aspects. It focuses on different values of different microblogs and removes the errors from garbage users. Our content based method breaks traditional limit of sentiment classification and based on a new box office oriented semantic classification, which is more relative with box offices. Meanwhile, the relationship between count based feature and content based feature is observed. New feature extraction formulas are built to mix these features. With these features, not only linear prediction model but also neutral network model and support vector model are selected to predict the box offices of certain movies.

The performances of our methods are evaluated from various aspects. Compared with not only previous microblog prediction methods but also traditional prediction methods, our methods are more accurate and reliable. According to our analysis of the results, the predictive power of microblog is different when it deals with different problems. When it predicts the outcomes which are relative the users' behaviors on social media, such as small ware sales volume and box offices, the performances will be better. On the contrary, it cannot have good performances on the problems like predicting the result of one election.

According to our work, there are lots of work to do in the future. Firstly, a method to recognize what kind of outcomes can be predicted by microblog prediction methods should be found. This is important to apply microblog prediction methods to real problems. And then, this paper provided methods to predict data in real world but microblog may be able to predict the events in real life. Methods which can solve this kind of problems are required. Besides, the methods can be improved by using the relationship among users. The features of complex social networks may make our prediction methods more accurate and flexible. Besides, the idea of latent semantic analysis (LSA) can be applied to our box office oriented classification method.

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